Reg. No: .....

Name: .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) EXAMINATION

FIFTH SEMESTER

Skill Enhancement Course (SEC) – MG5SECZGY301: AQUARIUM FABRICATION AND SETTING

(For those opting Aquaculture as Minor)

Duration: 1.5 Hours

Maximum Marks: 50

Learning Domains: Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S)

Students should attempt questions ensuring coverage of all Course Outcomes (CO1 to CO4).

Part A (Multiple Choice Questions)

Answer all questions. Each question carries 1 mark.

1.Which of the following materials offers superior clarity and seamless curved edges for custom tank designs? [K] / [CO1]

a) Annealed Float Glass b) Cast Acrylic Sheet

c) Tempered Glass d) Polycarbonate Sheet

2. A Paludarium is a specialized aquarium system designed to integrate: [K] / [CO1]

a) Marine corals and deep-water macroalgae b) Both terrestrial (land) and aquatic semi-submerged environments c) Low-temperature coldwater stenohaline organisms

d) High-turbidity brackish mangrove mud substrates

3.Which filtration component is exclusively used in saltwater marine systems to remove dissolved organic proteins via micro-foaming? [K] / [CO2]

a) Hang-On-Back (HOB) Filter b) Sponge Filter

c) Under Gravel Filter (UGF) d) Protein Skimmer

4.In an established aquarium biological filter, *Nitrosomonas* bacteria primarily oxidize: [U] / [CO2]

a) Nitrate to Nitrogen gas b) Toxic Ammonia into Nitrite

c) Nitrite into Nitrate d) Molecular Oxygen into Carbon Dioxide

5.Ceramic rings and porous sintered glass media in a canister filter serve primarily as: [K] / [CO2]

a) Chemical adsorbents b) Mechanical strainers

c) Biological nitrifying substrate d) Water-softening resins

6.Which lighting spectrum and type is most energy-efficient and widely preferred for modern high-tech planted aquariums? [K] / [CO2]

a) Incandescent tungsten bulbs b) Metal Halide fixtures

c) Full-spectrum LED systems d) Standard T12 fluorescent tubes

7.In the placement hierarchy of aquascaping layout design, *Vallisneria spiralis* is typically classified as a: [U] / [CO3]

a) Foreground carpet plant b) Midground accent plant

c) Background tall plant d) Surface floating bio-cover

8.Which live food organism is most suitable as the first feed for tiny newly hatched fish fry that cannot consume Artemia nauplii? [K] / [CO3]

a) *Tubifex* worms b) Bloodworms (*Chironomus* larvae)

c) Infusoria (Ciliates) d) Adult *Daphnia magna*

9.The process of gradually equalizing water temperature and chemistry before releasing new fish into a display tank is known as: [U] / [CO3]

a) Nitrogen Cycling b) Acclimatization

c) Biological Conditioning d) Prophylactic Dip

10.What is the fundamental cause of "New Tank Syndrome" in newly fabricated, uncycled aquaria? [U] / [CO4]

a) Depletion of trace minerals b) Accumulation of toxic Ammonia and Nitrite due to absence of nitrifiers c) Excessive proliferation of green hair algae d) Sudden depletion of carbonate hardness (KH)

11.White Spot Disease (Ich) in freshwater tropical fish is caused by the ciliate protozoan: [K] / [CO4]

a) *Ichthyophthirius multifiliis* b) *Saprolegnia parasitica*

c) *Aeromonas hydrophila* d) *Dactylogyrus vastator*

12.Conditioning ornamental fishes prior to long-distance transportation involves: [U] / [CO4]

a) Heavy feeding with high-protein live worms b) Starvation for 24–48 hours to minimize metabolic waste excretion c) Raising the water temperature by 5°C d) Adding excessive antibiotic doses into shipping bags

1x12 = 12

Part B (Short Answer Questions)

Answer any 6 out of 8 questions. Each question carries 3 marks.

13. Compare glass and acrylic as tank construction materials based on optical clarity, scratch resistance, and structural adaptability. [U] / [CO1]
14. Distinguish between a Fish-Only with Live Rock (FOWLR) tank and a specialized Reef Tank. [U] / [CO1]

15. Outline the working mechanism and utility of a Hang-On-Back (HOB) filter in home aquaria. [U] / [CO2]
16. Describe the biochemical steps involved in the aquarium Nitrogen Cycle. [U] / [CO2]
17. Explain the standard stepwise protocol for introducing newly purchased fish into a seasoned aquarium. [A] / [CO3]
18. Outline the standardized laboratory/home culture procedure for *Daphnia* or *Moina*. [S] / [CO3]
19. State the major causative factors for excessive algae blooms in freshwater tanks and suggest three preventive measures. [U] / [CO4]
20. Explain the essential packaging steps followed during the live transport of ornamental fishes in sealed polythene bags. [A] / [CO4]

6x3= 18

Part C (Short Essay Questions)

Answer any 4 out of 6 questions. Each question carries 5 marks.

21. Classify aquariums based on water salinity and design concepts (Freshwater community, Biotope, Nano, Paludarium, and Reef systems). [U] / [CO1]
22. Provide a detailed account of essential water quality parameters pH, Dissolved Oxygen, Ammonia, Nitrite, Nitrate, and Hardness) and their acceptable ranges for tropical fish maintenance. [U] / [CO2]
23. Detail the stepwise construction and fabrication of an all-glass rectangular freshwater aquarium, including glass cutting, silicone application, curing, and leak-testing. [A, U] / [CO3]
24. Classify aquatic plants based on their placement zones in aqua scaping (foreground, midground, background, and floating) and explain their physiological role in biological balance. [U] / [CO3]
25. Describe the etiology, symptoms, and therapeutic treatment protocols for one bacterial and one fungal disease commonly encountered in tropical ornamental fishes. [U, A] / [CO4]
26. Discuss the commercial aspects, conditioning strategies, oxygenation methods, and modern packing technologies used in the ornamental fish export and domestic marketing sector. [U, A] / [CO4]

4x5=20

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